

Problem Set 1

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1 Basic logic

Problem 1: Find the truth tables for each of the following propositions:

- $A \vee \neg A$
- $A \Leftrightarrow B$
- $(A \vee \neg B) \wedge (A \Rightarrow \neg C)$

Problem 2: Let the notation $\exists!$ denote “unique existence”, so for instance, $\exists!x \in A, R(x)$ means that there is only one x so that $R(x)$ is True. Show how we can express unique existence using “standard” logical operators – that is, show how to express “ $\exists!x \in A, R(x)$ ” only using $\wedge, \vee, \neg, =, \exists$.

Problem 3: Negations

Find the negations of the following propositions. Express your answers using only $\vee, \wedge, P, Q, R, S, \neg P, \neg Q, \neg R, \neg S$.

1. $P \Rightarrow Q$
2. $(P \vee \neg Q) \wedge (R \vee \neg S)$
3. $(P \Rightarrow Q) \Rightarrow \neg R$

2 Set theory

Problem 4: Let $S_1 = \{Red, Blue\}$, $S_2 = \{1, 2\}$, and $S_3 = \{2, 3\}$. Compute the following:

- $S_1 \times (S_2 \cup S_3)$
- $S_1 \cap S_2$
- $\{orange\} \times (S_2 \cap S_3)$

Problem 5: Let S_1, S_2, S_3 be sets. Recall that set difference, denoted $A \setminus B$ is defined to be the set of all elements in set A that are not in set B . Prove that $(S_1 \setminus S_2) \cup (S_2 \setminus S_3) \subseteq (S_1 \cup S_2) \setminus (S_1 \cap S_2 \cap S_3)$.

Problem 6a: Let A, B be finite sets so that $|A| = |B| = n$. How many bijections are there that map from $A \rightarrow B$? Recall that a bijection is a map that is both injective and surjective. Explain your answer (i.e., demonstrate the bijection explicitly).

Problem 6b: Let A, B be finite sets so that $|A| = n$ and $|B| = m$. How many functions are there that map from $A \rightarrow B$? Once more, explain your answer.